

Meridian Supply Chain RAG Assistant

A Retrieval-Augmented Generation assistant that lets a buyer ask plain-English questions and get answers drawn from Meridian Components' Q1 FY2025-26 Supply Chain Review and its Procurement Policy Handbook v4.2 — with the source document and page shown under every answer.

Uses the **Gemini API** for both embeddings and answering, with OpenAI available as an optional fallback if a key is configured.

Demo Video

~3 minutes — upload, indexing, cross-document questions, and the trap question being refused.

- [▶ Watch on Google Drive](#)

Architecture

```
PDF -> pdfplumber (per-page text extraction)
      -> recursive character chunking (1100 chars, 150 char overlap)
      -> Gemini embedding (gemini-embedding-001, 768-dim), batched
      -> ChromaDB (persisted to ./chroma_db, deterministic chunk ids so
          re-indexing upserts instead of duplicating)
      -> [on a question] embed question -> Chroma similarity search
          (top_k, default 8, adjustable via a slider in the UI)
      -> Gemini generation, tried across a short model fallback list (see
          GEMINI_CHAT_MODELS in llm_providers.py)
      -> Streamlit UI displays the answer and the source file/page for
          every chunk that was retrieved
```

Orchestration uses the plain `google-genai` SDK directly (no LangChain/ LangGraph for the RAG logic itself — `langchain-text-splitters` is used only for the chunking step), kept deliberately simple per the assignment's "keep it simple" guidance.

Why a model fallback chain, and how it works

Gemini has been retiring model IDs faster than their published shutdown dates, so `llm_providers.py` doesn't hardcode a single model name for answering. `ask_llm()`:

1. Tries each model in `GEMINI_CHAT_MODELS` in order — currently `gemini-3.6-flash`, then `gemini-3.5-flash-lite` — moving to the next one if a call raises any exception (including a rate-limit or a retired-model error).
2. Falls back to OpenAI's `gpt-4o` if `OPENAI_API_KEY` is set and every Gemini attempt failed.
3. Raises with the collected error from every attempt only if all configured providers fail.

This is a straightforward try-next-on-failure chain, not a retry-with- backoff loop — a transient error on a given model moves straight to the next model rather than retrying the same one. That's a reasonable trade-off for a

project this size, but it does mean a single flaky response can burn through the whole list faster than it needs to; adding one retry with a short backoff before falling through would be a natural next improvement.

Embeddings only have one current Gemini text-embedding model (`gemini-embedding-001`), so there's no fallback model on that side; the 768-dimension output (Matryoshka-truncated from the 3072-dim default) keeps the Chroma index small with negligible quality loss for chunks this short.

Setup

1. Clone the repo and `cd` into it.
2. Create a virtual environment and install dependencies:

```
python3 -m venv .venv
source .venv/bin/activate      # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

3. Copy `.env.example` to `.env` and add your Gemini API key (free at <https://aistudio.google.com/apikey>):

```
cp .env.example .env
# edit .env: GEMINI_API_KEY=AIza...
```

An `OPENAI_API_KEY` can also be added as a fallback for answering only (not embeddings) — leave it blank if you don't have one.

4. Run the app:

```
streamlit run app.py
```

5. In the sidebar, upload the two PDFs from `data/` (or your own), click **Index documents**, then ask a question in the main panel.

Indexed data persists in `./chroma_db` — closing and reopening the app does not require re-uploading, as long as `chroma_db/` still exists on disk.

Optional: FastAPI backend

```
uvicorn api.main:app --reload
```

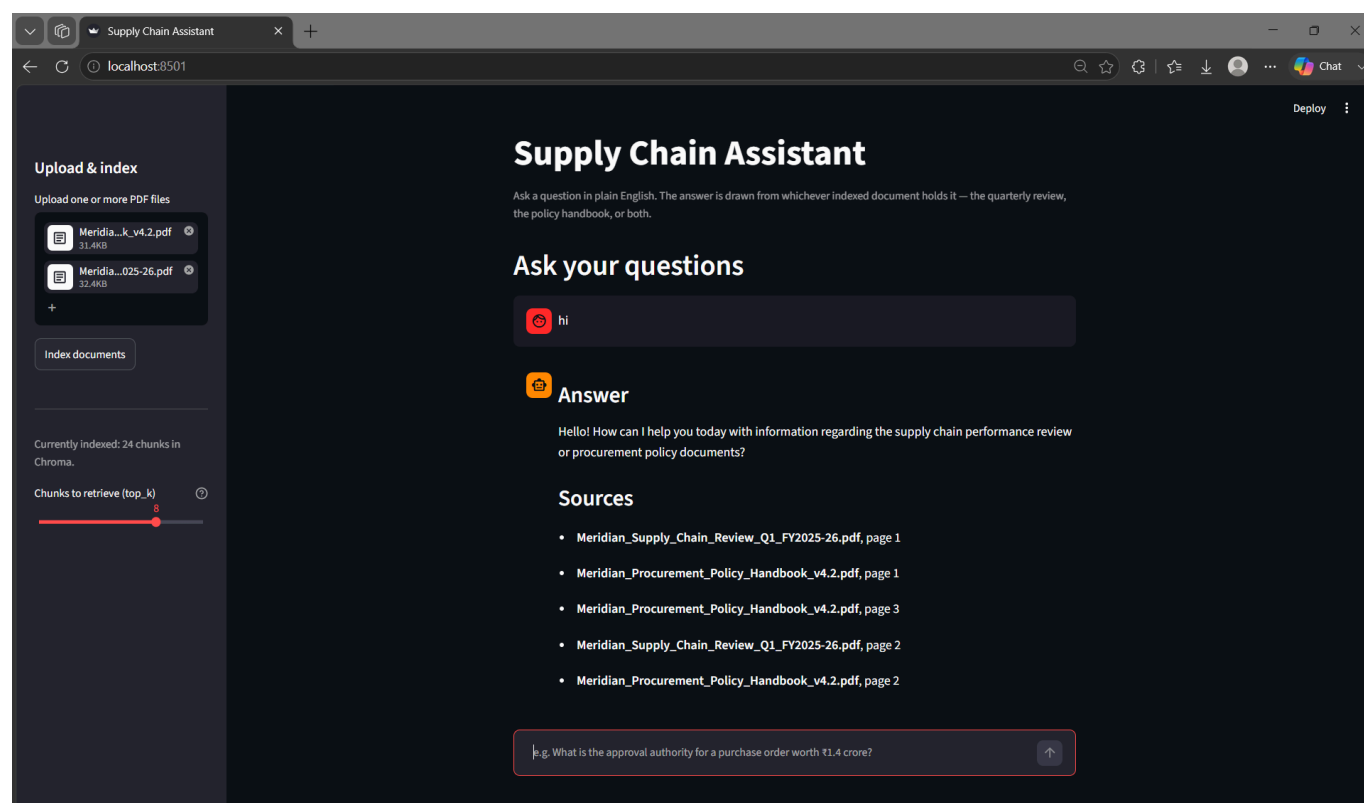
Then open `http://localhost:8000/docs` to try `/ingest`, `/ask`, and `/stats` directly. Live responses from this project's own run are in `screenshots/fastapi/` (`HOME.png`, `INGEST.png`, `ASK-1.png`, `ASK-2.png`, `STATS.png`) and logged in `screenshots/FASTAPI.txt`.

Chunking choice

Chunk size 1100 characters, 150 character overlap. Both PDFs are table-heavy — supplier scorecards, freight lane tables, approval-authority tables — and a chunk near the top of the assignment's 800–1200 range keeps a full table row, or a row plus its header, together rather than splitting a number away from the label that explains it. The 150-char overlap ensures a sentence sitting on a chunk boundary is still whole in at least one chunk.

Actual result on the two provided PDFs, verified by running the load/chunk step standalone: **24 chunks total** — **11 from the Supply Chain Review, 13 from the Procurement Policy Handbook** (confirmed by [screenshots/FASTAPI.txt](#), which shows `"total_chunks": 24` from a live `/ingest` call).

Landing Page UI



Test questions and answers

All ten answers below were generated by the running app against the indexed PDFs (full text also saved in [screenshots/answers.txt](#)), with a screenshot from the actual run for each one.

1. Which supplier had the highest spend in Q1, and what was its on-time delivery percentage?

Shenzhen Rui Electronics had the highest Q1 spend, at ₹21.9 crore, with an on-time delivery rate of

79.5%. (Source: *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, pages 1–3)

 Which supplier had the highest spend in Q1, and what was its on-time delivery percentage?

 **Answer**

Based on *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1 (Section 2, *Supplier performance scorecard* and *Notes on the scorecard*):

- **Supplier with the highest spend:** Shenzhen Rui Electronics (with a Q1 spend of ₹21.9 crore)
- **On-time delivery percentage:** 79.5%

Sources

- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 3
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2

2. How many line stoppages happened in Q1, what was the total downtime, and what caused them?

7 line-stoppage events across two plants, 41 hours of total downtime. Causes: a microcontroller shortage from Shenzhen Rui Electronics (4 events, 22 hrs — a vessel roll-over, a 9-day customs hold, a partial shipment, and an allocation shortfall), PCB quality rejections from Trident Circuit Boards (2 events, 11 hrs), and a transporter strike on the Coimbatore–Pune corridor (1 event, 5 hrs). (Sources: *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, pages 1–3)



How many line stoppages happened in Q1, what was the total downtime, and what caused them?



Answer

According to *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf* (Page 1 and Page 2):

- **Number of Line Stoppages:** 7 line-stoppage events were recorded across the two plants (Page 1, Section 1 and Page 2, Section 5).
- **Total Downtime:** 41 hours (Page 1, Section 1 and Page 2, Section 5).
- **Causes:**
 1. **Microcontroller Shortage from Shenzhen Rui Electronics** (4 events, 22 hours total) (Page 1, Section 1 and Page 2, Section 5):
 - 22 Apr 2025 (Chakan, 4 hrs): Vessel roll-over at Shenzhen.
 - 09 May 2025 (Chakan, 11 hrs): 9-day customs hold at Nhava Sheva.
 - 27 May 2025 (Chakan, 6 hrs): Partial shipment received.
 - 25 Jun 2025 (Chakan, 4 hrs): Allocation shortfall from supplier.
 2. **PCB Quality Issues from Trident Circuit Boards** (2 events, 11 hours total) (Page 2, Section 5):
 - 14 May 2025 (Hosur, 3 hrs): PCB lot rejected at incoming inspection.
 - 17 Jun 2025 (Hosur, 8 hrs): PCB lot rejected at incoming inspection.

3. **External Transport Disruption** (1 event, 5 hours total) (Page 2, Section 5):

- 06 Jun 2025 (Chakan, 5 hrs): Transporter strike on the Coimbatore–Pune corridor.

Sources

- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 3

3. **What is the approval authority for a purchase order worth ₹1.4 crore?** Per Section 3 of the handbook, an order above ₹1 crore and up to ₹5 crore requires Chief Operating Officer approval — so ₹1.4 crore needs COO sign-off. (Source: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, pages 1–3)



What is the approval authority for a purchase order worth ₹1.4 crore?



Answer

Based on Section 3 ("Approval authority for purchase orders") of *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 1:

- For a purchase order value **above ₹1 crore and up to ₹5 crore**, the approving authority is the **Chief Operating Officer**.

Since ₹1.4 crore falls within this bracket, the approval authority for a ₹1.4 crore purchase order is the **Chief Operating Officer**.

Sources

- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 1
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 3

4. **What are the four supplier classification categories, and what qualifies a supplier as Critical?** The four categories are Critical, Strategic, Standard, and Tail. A supplier qualifies as Critical if it meets any one of: single-source for any part, annual spend above ₹10 crore, or supplies a safety-related component. (Sources: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, pages 1–2; *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1)

 What are the four supplier classification categories, and what qualifies a supplier as Critical?

 **Answer**

According to *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 1, Section 2 ("Supplier classification"):

Four Supplier Classification Categories:

1. Critical
2. Strategic
3. Standard
4. Tail

Qualifying Criteria for a "Critical" Supplier:

Meeting any one of the following criteria qualifies a supplier as Critical:

- Single-source for any part
- Annual spend above ₹10 crore
- Supplies a safety-related component

Sources

- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 1
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1

5. **Kaveri Metals recorded 88.1% on-time delivery and 1,150 defects per million in Q1. Which policy clauses does this trigger, and what exactly must the buyer do?** Triggers Clause 6.1 (on-time delivery below 90%) and Clause 6.3 (defect rate above 500 PPM). Under 6.1: issue a written warning within 10 working days of quarter close and move the supplier to weekly delivery review calls until it recovers above 90% for a full quarter. Under 6.3: recover rework cost at ₹120 per affected unit and impose 100% incoming inspection at the supplier's cost until three consecutive clean lots. (Sources: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2; *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, pages 1–3)



Kaveri Metals recorded 88.1% on-time delivery and 1,150 defects per million in Q1. Which policy clauses does this trigger, and what exactly must the buyer do?



Answer

Based on the provided documents, Kaveri Metals' performance in Q1 (88.1% on-time delivery and 1,150 PPM defect rate) triggers **Clause 6.1** and **Clause 6.3** of the *Meridian Procurement Policy Handbook v4.2.pdf* (page 2):

1. Triggered Clauses & Conditions:

- **Clause 6.1 (On-time delivery below 90% in any quarter):**
 - *Condition:* Kaveri Metals' on-time delivery of 88.1% is below the 90% threshold for Q1 [Source: *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1].
- **Clause 6.3 (Defect rate above 500 parts per million in any quarter):**
 - *Condition:* Kaveri Metals' defect rate of 1,150 PPM exceeds the 500 PPM threshold [Source: *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1].

(Note: Clause 6.2 is not triggered as OTD is not below 85% for two consecutive quarters).

2. Actions Required / Consequences:

- **Under Clause 6.1** [Source: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2]:
 - Issue a written warning to Kaveri Metals within **10 working days** of quarter close.
 - Move Kaveri Metals to a **weekly delivery review call** until performance recovers above 90% for one full quarter.
- **Under Clause 6.3** [Source: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2]:
 - Recover the cost of rework from Kaveri Metals at a standard recovery rate of **₹120 per affected unit**.
 - Impose **100% incoming inspection** at Kaveri Metals' cost until three consecutive lots are accepted without defect.

Sources

- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 3

6. **The microcontroller supplier is single-source. What does the sourcing policy require in this situation, and what is the company already doing about it?** Being single-source makes Shenzhen Rui Electronics a Critical supplier (Section 2). Section 7.1 requires a qualified second source within 12 months of that classification, with monthly progress reported to the Management Committee; Section 7.2 caps any single supplier at 60% of a part's volume once dual-sourced, absent written COO approval. The company is already qualifying an alternate supplier. (Sources: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, pages 1–3; *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1)



The microcontroller supplier is single-source. What does the sourcing policy require in this situation, and what is the company already doing about it?



Answer

Based on the provided documents, here are the classification requirements, sourcing policy rules, and the actions currently being taken:

1. Classification and Sourcing Policy Requirements

- **Classification:** Under Section 2 of the *Meridian Procurement Policy Handbook v4.2.pdf* (page 1), a supplier meeting any criteria such as being a single-source for any part, having annual spend above ₹10 crore, or supplying safety-related components is classified as **Critical** (otherwise, suppliers are classified as Strategic, Standard, or Tail depending on spend). Because Shenzhen Rui Electronics is the single source for microcontrollers (*Meridian Supply Chain Review Q1 FY2025-26.pdf*, page 1), it falls under the **Critical** supplier classification.
- **Dual Sourcing Requirement:** Section 7.1 of the *Meridian Procurement Policy Handbook v4.2.pdf* (page 2) requires that every part supplied by a Critical supplier must have a qualified second source in place within 12 months of that classification being assigned, with monthly progress reported to the Management Committee.
- **Volume Share Limit:** Section 7.2 of the *Meridian Procurement Policy Handbook v4.2.pdf* (page 2) mandates that once dual sourcing is established, no single supplier may hold more than 60% of the volume of any one part without written COO approval.

2. What the Company is Doing

- **Qualification of Alternate Supplier:** Qualification of an alternate supplier is already under way (*Meridian Supply Chain Review Q1 FY2025-26.pdf*, page 1).

Sources

- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 3
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 1
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 3

7. **Microcontrollers are imported with a 46-day lead time. Using the safety-stock policy, how many days of stock should be held for this part?** Base formula: $46 \times 0.25 = 11.5$ days. The minimum floor for an imported, Critical-supplier part is 30 days, which is higher than the calculated value — and the microcontroller supplier is Critical (single-source, currently undergoing dual-sourcing). So 30 days of stock should be held. (Sources: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, pages 2–3; *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, pages 1–3)



Microcontrollers are imported with a 46-day lead time. Using the safety-stock policy, how many days of stock should be held for this part?



Answer

Based on the **Meridian Procurement Policy Handbook v4.2 (page 3)**, safety stock is calculated as: $\text{Safety stock (days)} = \text{Lead time in days} \times 0.25 = 46 \times 0.25 = 11.5 \text{ days}$

The policy requires comparing the calculated value to the minimum floor and applying the higher value:

1. **If classified as Imported, non-critical supplier:** The minimum floor is **10 days** [Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3]. The higher value is **11.5 days**.
2. **If classified as Imported, Critical supplier:** The minimum floor is **30 days** [Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3]. The higher value is **30 days**.

Evidence & Conclusion: Microcontrollers are imported from Shenzhen Rui Electronics, which is currently a single source undergoing dual-sourcing qualification under Section 7.1 rules for Critical suppliers [Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 2] and is identified as the highest-rated supply risk [Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 1, 3]. Because it is supplied by a Critical supplier, the 30-day minimum floor applies as it is higher than the calculated 11.5 days.

Final Answer: 30 days of stock should be held [Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3].

Sources

- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 2
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 1
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 3
- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 2

8. **Trident Circuit Boards had a defect rate of 640 parts per million. What is the cost consequence under the policy?** 640 PPM exceeds the 500 PPM threshold, triggering Clause 6.3: the supplier bears rework cost at ₹120 per affected unit, and 100% incoming inspection is imposed at the supplier's cost until three consecutive lots are accepted defect-free. (Sources: Meridian_Procurement_Policy_Handbook_v4.2.pdf, pages 2–3; Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, pages 1–2)



Trident Circuit Boards had a defect rate of 640 parts per million. What is the cost consequence under the policy?



Answer

Based on the provided documents, the policy consequences depend on whether a supplier's defect rate is above or below 500 parts per million (PPM) in any quarter:

- If the defect rate is 500 PPM or below: Clause 6.3 does not trigger.
- If the defect rate is above 500 PPM: Clause 6.3 of *Meridian_Procurement_Policy_Handbook_v4.2.pdf* (page 2) applies.

Evidence & Conclusion: According to *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf* (page 1 and page 2), Trident Circuit Boards had a defect rate of **640 PPM**, which satisfies the condition of being above 500 PPM.

Therefore, under **Clause 6.3** (*Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2), the cost consequences are:

1. The supplier bears the cost of rework at a standard recovery rate of **₹120 per affected unit**.
2. **100% incoming inspection** is imposed at the **supplier's cost** until three consecutive lots are accepted without defect.

Sources

- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 2
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 2
- *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, page 1
- *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, page 3

9. **Which suppliers would fall below the B rating band on on-time delivery alone, and what is the escalation path for them?** No supplier fell below Band B (below 75%) — the lowest was Shenzhen Rui Electronics at 79.5%. Three suppliers fell below Band A (below 90%): Shenzhen Rui Electronics (79.5%), Trident Circuit Boards (84.6%), and Kaveri Metals (88.1%), all triggering Clause 6.1. Shenzhen Rui Electronics additionally triggers Clause 6.2 (below 85% for two consecutive quarters: 83.2% in Q4 FY24-25, 79.5% in Q1 FY25-26) — a 2% debit note on quarterly invoice value plus a formal improvement plan within 15 working days, escalating to a business hold (Clause 6.4) if not submitted. The general escalation matrix (Section 10) runs Buyer (Level 1, ≤3-day slippage) → Category Manager (Level 2, >3-day slippage or a rejected lot) → Head of Procurement (Level 3, line-stoppage risk within 7 days) → COO (Level 4, actual stoppage or supplier insolvency signal). (Sources: *Meridian_Procurement_Policy_Handbook_v4.2.pdf*, pages 2–3; *Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf*, pages 1, 3)



Which suppliers would fall below the B rating band on on-time delivery alone, and what is the escalation path for them?



Answer

Based on the provided documents, here is the breakdown regarding rating bands on on-time delivery and their consequences/escalation paths:

1. Classification & Supplier Identification

According to [Meridian_Procurement_Policy_Handbook_v4.2.pdf](#) (page 2):

- A supplier delivering below 90% cannot score in **Band A**.
- A supplier delivering below 75% cannot score in **Band B** (meaning they fall below Band B into Band C or D).

Looking at the Q1 FY 2025-26 delivery performance from [Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf](#) (page 1):

- **Suppliers below 75% (falling strictly below Band B):** None. The lowest delivery performance was Shenzhen Rui Electronics at 79.5%.
- **Suppliers below 90% (falling below Band A into Band B or lower):**
 1. **Shenzhen Rui Electronics:** 79.5%
 2. **Trident Circuit Boards:** 84.6%
 3. **Kaveri Metals Pvt Ltd:** 88.1%

2. Consequences and Escalation Path

A. Policy Consequences for On-Time Delivery Failures ([Meridian_Procurement_Policy_Handbook_v4.2.pdf](#), page 2)

- **Delivery below 90% in any quarter (Clause 6.1):** Applies to Kaveri Metals, Trident Circuit Boards, and Shenzhen Rui Electronics.
 - A written warning is issued within 10 working days of quarter close.
 - The supplier moves to a weekly delivery review call until performance recovers above 90% for one full quarter.
- **Delivery below 85% for two consecutive quarters (Clause 6.2):** Applies specifically to Shenzhen Rui Electronics (Q4 FY 2024-25 at 83.2% and Q1 FY 2025-26 at 79.5%; [Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf](#), page 1).
 - In addition to Clause 6.1, a debit note equal to 2% of the quarterly invoice value is raised.
 - The supplier must submit a formal improvement plan within 15 working days. Failure to submit escalates to Clause 6.4 (business hold).

B. Escalation Matrix (Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3, Section 10)

If delivery slippages trigger formal escalations:

- **Level 1 (Buyer, 24-hour response time):** Triggered by delivery slippage of up to 3 days.
- **Level 2 (Category Manager, 48-hour response time):** Triggered by delivery slippage beyond 3 days, or a rejected lot.
- **Level 3 (Head of Procurement, 72-hour response time):** Triggered by risk of line stoppage within 7 days.
- **Level 4 (Chief Operating Officer, 5 working days response time):** Triggered by actual line stoppage or supplier insolvency signal.

Sources

- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 2
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 1
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 3
- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3

10. **Trap question** — "What is the annual salary of the Head of Procurement?" The information is not available in the uploaded documents — correctly refused rather than answered with an invented number. (Sources: Meridian_Procurement_Policy_Handbook_v4.2.pdf, pages 1, 3; Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 1)



What is the annual salary of the Head of Procurement?



Answer

Based on the provided documents, the information regarding the annual salary of the Head of Procurement is not available.

Sources

- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 1
- Meridian_Procurement_Policy_Handbook_v4.2.pdf, page 3
- Meridian_Supply_Chain_Review_Q1_FY2025-26.pdf, page 1

Known limitations / honest notes

- **All ten test questions were answered correctly** in this run, including both trap-style checks: Q9 correctly reported that *no* supplier fell below Band B rather than forcing a match, and Q10's salary question was correctly refused instead of guessed.
- **top_k matters more on the cross-document questions (5–9).** These need a number from the review *and* a clause from the handbook to land in the same context window; the code comment in `rag.py` calls out that `top_k` around 5–6 or higher is needed rather than 3–4, which is why the default was set to

8 and left adjustable via a slider in the UI rather than hardcoded. This run used the default of 8 throughout; the effect of lower values on these specific questions wasn't separately re-tested and logged for this README.

- **No retry-with-backoff on the answering call.** As noted in the Architecture section, `ask_llm()` moves to the next model in the list on any failure rather than retrying the same model with backoff first. On Gemini's free tier this means a single rate-limit hit skips straight to the next model instead of waiting it out, which is fine for this project's scale but wouldn't be efficient at higher volume.
- **The UI doesn't currently surface which model produced a given answer.** `ask_llm()` returns only the text, not which entry in `GEMINI_CHAT_MODELS` succeeded, so `app.py` has no "answered by" field to display. All ten answers above were produced without hitting the OpenAI fallback (Gemini answered successfully every time in this run).
- **Table extraction via `pdfplumber`** occasionally flattens multi-column scorecard rows into a single line without clear column boundaries; the 1100-character chunk size was chosen specifically to make this less of a problem, and it held up correctly across all ten test questions here, but a table-aware parser would be a more robust fix for larger or messier source documents.

Repository structure

```
supplychain-rag/
├─ app.py           # Streamlit interface
├─ ingest.py        # load, chunk, embed, store in Chroma
├─ rag.py           # retrieve + prompt + call the answering model
├─ llm_providers.py # embedding/answering provider abstraction (Gemini +
OpenAI)
├─ api/main.py      # optional FastAPI backend
├─ data/            # the two provided PDFs
├─ screenshots/     # app + FastAPI screenshots, recorded answers
├─ chroma_db/       # persisted vector store (gitignored)
├─ video/
│   └─ demo.webm    # 3-minute demo video
├─ requirements.txt
├─ .env.example
├─ .gitignore
└─ README.md
```